

**GUJARAT TECHNOLOGICAL UNIVERSITY****BE- SEMESTER-I & II EXAMINATION – WINTER 2024****Subject Code:BE01000091****Date:09-01-2025****Subject Name:Mechanics of Solids****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

|            |                                                                                                                                                                                                                                                                                      | Marks     |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Q.1</b> | <b>(a)</b> What are the methods for finding out the resultant forces for a given system of forces ?                                                                                                                                                                                  | <b>03</b> |
|            | <b>(b)</b> State (1) parallelogram law of forces. and (2) two characteristics of a couple                                                                                                                                                                                            | <b>04</b> |
|            | <b>(c) Objective Question (MCQ)</b>                                                                                                                                                                                                                                                  | <b>07</b> |
|            | 1. In order to determine the effects of a force acting on a body we must know (a) Its magnitude and direction of the line along which it acts (b) Its nature (whether push or pull), (c) Point through which it acts on the body (d) All of the above                                |           |
|            | 2. The lami's theorem is applicable only for (a) coplanar forces, (b) concurrent forces, (c) coplanar & concurrent forces, (d) any type of forces                                                                                                                                    |           |
|            | 3. A body is said to be in equilibrium if it has no linear motion, (a) true , (b) False                                                                                                                                                                                              |           |
|            | 4. Lami's theorem cannot be applied in case of concurrent forces. (a) Agree, (b) Disagree.                                                                                                                                                                                           |           |
|            | 5. The shape of shear force diagram for cantilever beam subjected to couple at free end is (a) Horizontal straight line, (b) Zero, (c) Parabola, (d)Hexagonal                                                                                                                        |           |
|            | 6. The reaction at the roller support end of a beam is always (a) Vertical, (b) Horizontal, (c) None of the above                                                                                                                                                                    |           |
|            | 7. At free end of a cantilever bending moment is always_____ unless a concentrated moment is applied at the free end. (a) one (b) two (c) Three (d) Zero                                                                                                                             |           |
| <b>Q.2</b> | <b>(a)</b> Three concurrent coplanar forces act on a body at a point O, Determine two additional force along OA and OB such that resultant of the five forces is Zero as shown in <b>Fig-1</b>                                                                                       | <b>03</b> |
|            | <b>(b)</b> A boat is pulled along the river by two ropes with pulls P & Q inclined at 30 & 40 to the X-axis as shown in <b>Fig-2</b> find a) P and Q if their resultant R is 1000N parallel to X-axis b) if P is inclined at 30 to X-axis. find the minimum value of Q if R is same. | <b>04</b> |
|            | <b>(c)</b> A simply supported beam having a Maximum bending moment is 10 KN-m. the cross section of the beam is shown in <b>Fig-3</b> Determine bending stress in tension and compression.                                                                                           | <b>07</b> |
|            | <b>OR</b>                                                                                                                                                                                                                                                                            |           |
|            | <b>(c)</b> Explain (1) Types of Beam, Support, Load 2) Condition of Equilibrium                                                                                                                                                                                                      | <b>07</b> |
| <b>Q.3</b> | <b>(a)</b> Find analytically the support reaction at B for the beam shown in <b>Fig-4</b> if reaction at support A is zero                                                                                                                                                           | <b>03</b> |
|            | <b>(b)</b> Draw shear force, bending moment diagram for the beam shown in <b>Fig-5</b>                                                                                                                                                                                               | <b>04</b> |
|            | <b>(c)</b> A circular rod of 25mm diameter and 500mm long is subjected to a tensile force of 50 KN. Determine modulus of rigidity, bulk modulus and change in volume if poissions ratio is 0.3 and young's modulus E is $2 \times 10^5 \text{ N/mm}^2$                               | <b>07</b> |

**OR**

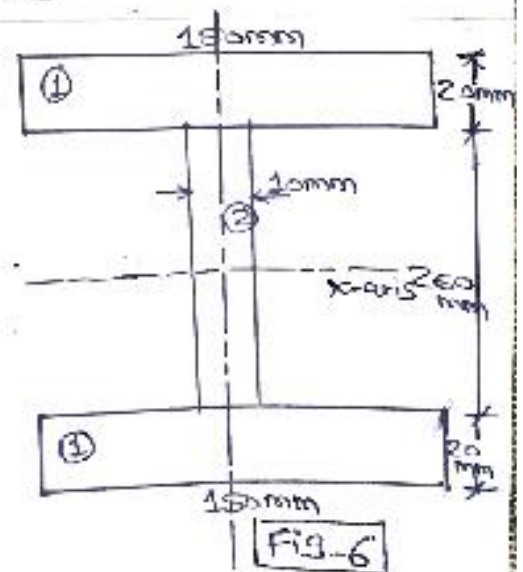
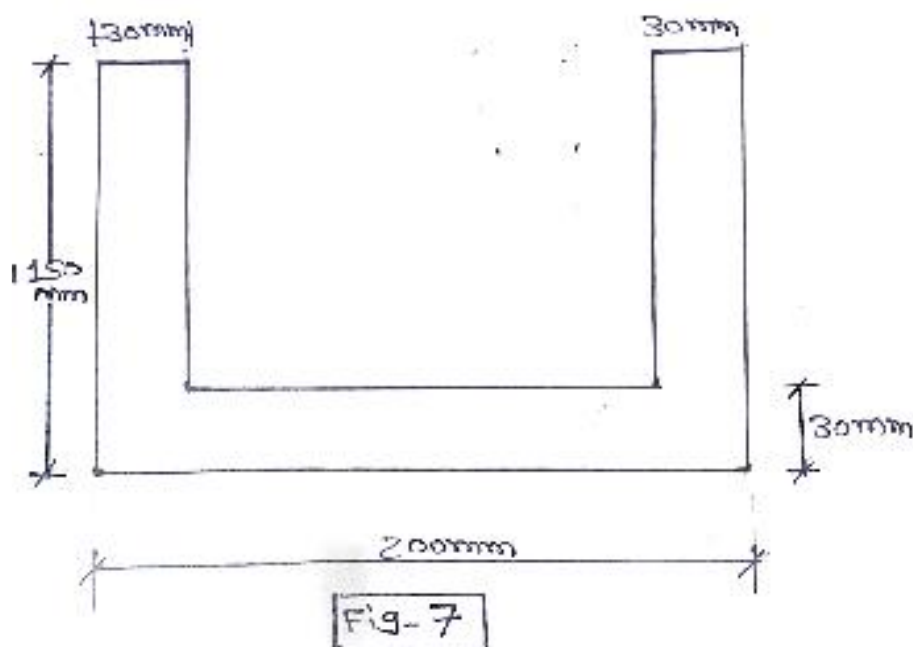
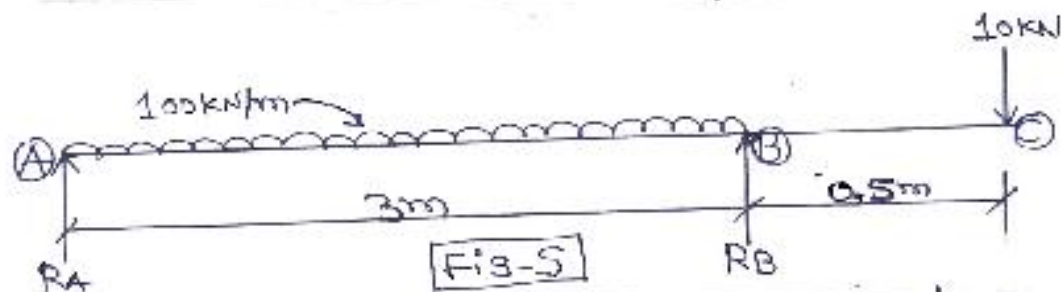
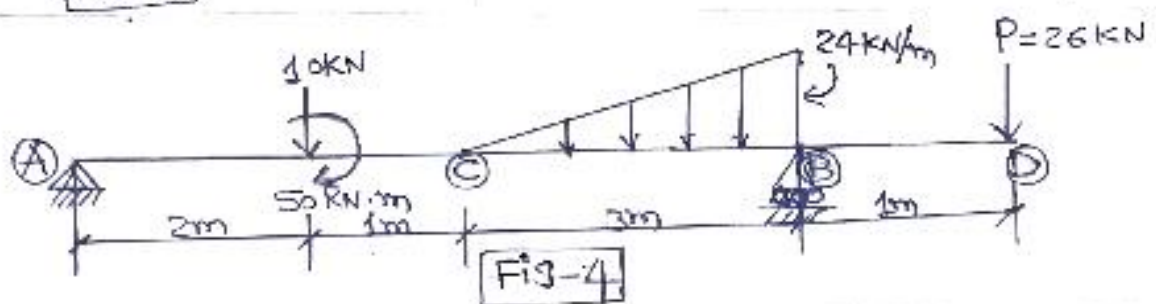
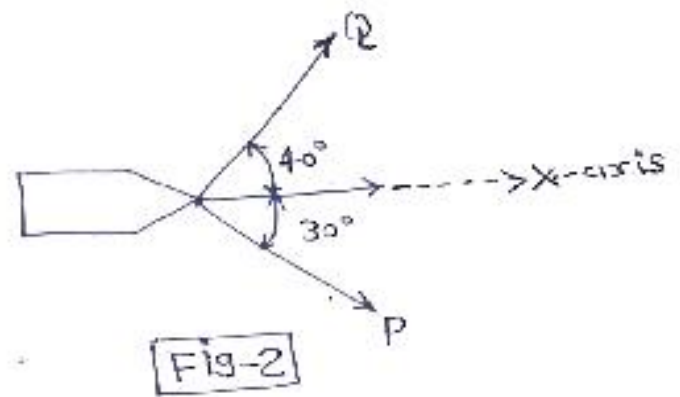
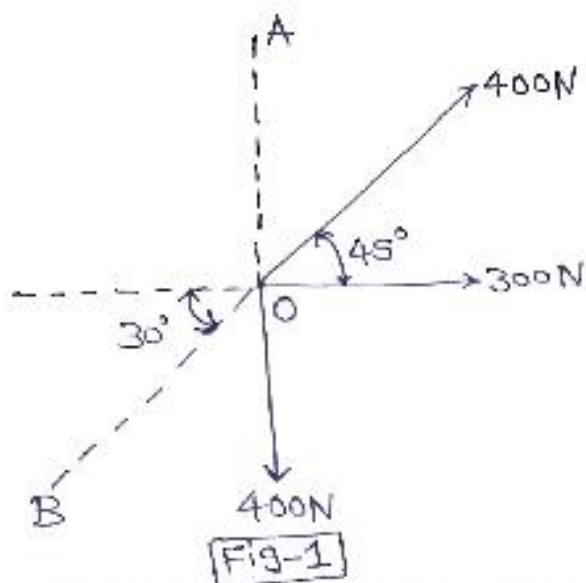
- Q.3** (a) Explain (a) Center of mass (b) center of gravity (c) centroid **03**  
(b) For an I-Section shown in **Fig-6** find  $I_{xx}$  and  $I_{yy}$  **04**  
(c) Find centroidal M.I  $I_{xx}$  and  $I_{yy}$  of a channel section shown in **Fig-7** **07**
- Q.4** (a) Write any three Assumption made in the theory of torsion **03**  
(b) A stepped bar made of steel, copper and brass are as under axial force as shown in **Fig-8** and is in equilibrium. The diameter of steel is 12 mm diameter of copper is 16mm and the diameter of brass is 20mm. determine total change in length of the bar take  $E_{\text{Steel}} = 200 \times 10^3 \text{ mpa}$   $E_{\text{Copper}} = 100 \times 10^3 \text{ mpa}$ ,  $E_{\text{Brass}} = 80 \times 10^3 \text{ mpa}$  **04**  
(c) The sum of two concurrent forces P and Q is 270 N and their resultant is of 180 N. if resultant is perpendicular to R. find P and Q **07**

**OR**

- Q.4** (a) How would you find out the moment of inertia of a plane area? **03**  
(b) Define the term (1) force (2) Couple Moment (3) Maximum Bending moment (4) Point of Contra flexure. **04**  
(c) Derive the equation of Bending Stress in Beam (Derivation of flexure Formula for pure Bending) **07**
- Q.5** (a) Short notes on (a) Moment of resistance (b) section modulus (c) Neutral Axis **03**  
(b) A beam of T-Section, 4m long carries a uniformly distributed load w per meter run through its length. The beam is simply supported at its ends the T-section is 20 x 10 x 1.2cm and web is 18.8 x 1.2 cm and flange is 10 x 1.2cm what is the maximum value of W so that the stress in the section does not exceed 60 mpa **04**  
(c) A simply supported beam 7m long is carrying three-point load 100N, 600N, and 175 N acting at 2m, 4m and 5.5 m from left support. Determine support reaction and draw S.F and B.M Diagram. **07**

**OR**

- Q.5** (a) Short notes on followings (1) Lateral strain (2) Hooks Law (3) Poisson's ratio **03**  
(b) Determine the stress, strain, modulus of elasticity and poissions ratio from the following results for a bar tested on UTM diameter 20mm, gauge length 150mm, increase in gauge length 14mm, decrease in diameter 0.8mm, tensile load 6 kN **04**  
(c) A rectangular bar of steel 50mm x 30 mm is 1m long carries a pull of 93.9 KN. Find the tensile stress in the bar strain and change in length. Take E 200 Gpa **07**



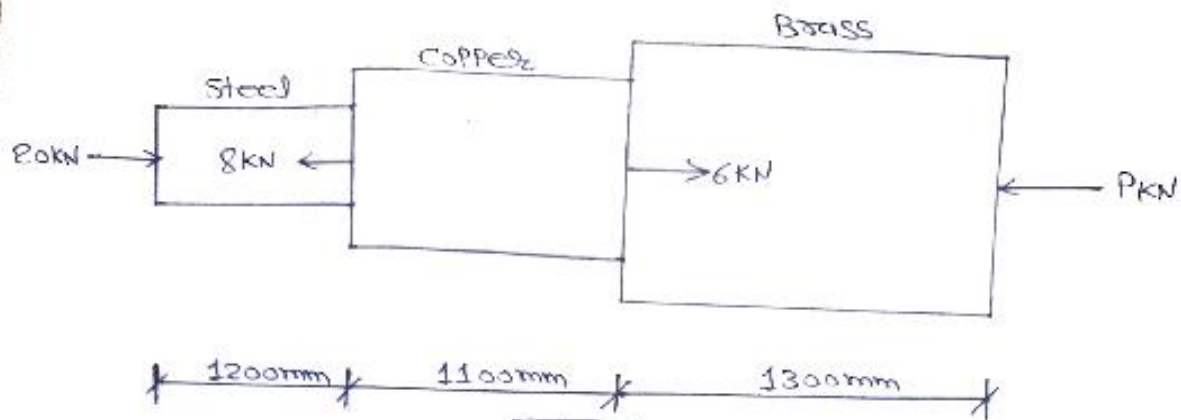


Fig-8

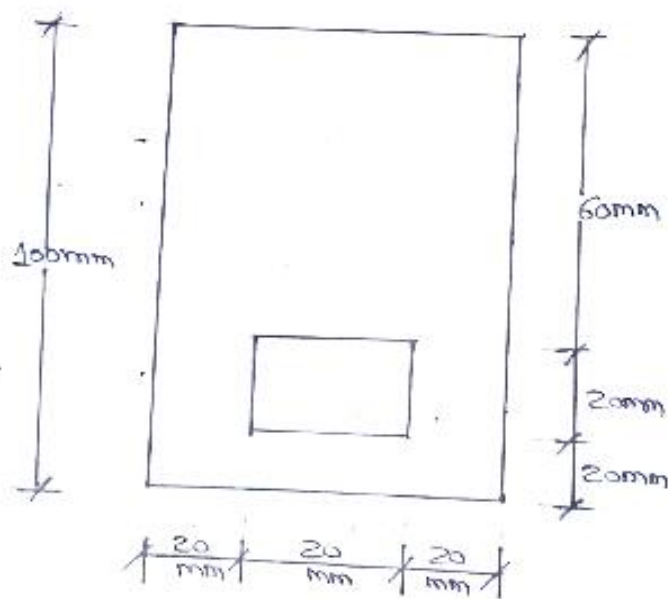


Fig-3

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# GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:BE01000091

Date:09-06-2025

Subject Name:Mechanics of Solids

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- |                                                                                                                                                          | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Q.1 (a) Define: (i) Rigid body, (ii) deformable body, (iii) Elastic body.                                                                                | 03    |
| (b) State and explain parallelogram law of forces.                                                                                                       | 04    |
| (c) Find resultant force in magnitude and direction if a particle is acted upon by following forces.                                                     | 07    |
| 1. 20 N inclined at 30 north of east.                                                                                                                    |       |
| 2. 25 N towards north                                                                                                                                    |       |
| 3. 30 N towards north-west                                                                                                                               |       |
| 4. 35 N inclined at 40 to south of west                                                                                                                  |       |
| 5. 24 N inclined at 30 to east of south.                                                                                                                 |       |
| Q.2 (a) State and explain Varignon's theorem.                                                                                                            | 03    |
| (b) Explain various types of load, beams and their support system.                                                                                       | 04    |
| (c) The beam AB of span 12 m shown in Fig. 1 is hinged at A and is on rollers at B. Determine the reactions at A and B for the loading shown in the Fig. | 07    |

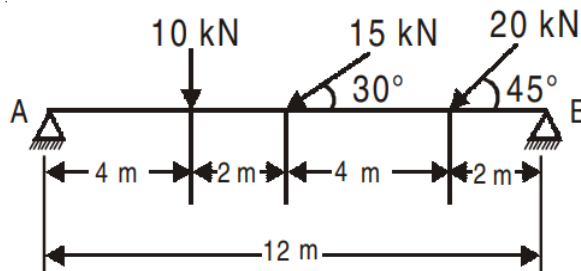


Fig. 1

OR

- |                                                                                         |    |
|-----------------------------------------------------------------------------------------|----|
| (c) Draw shear force and bending moment diagram for the cantilever beam shown in Fig. 2 | 07 |
|-----------------------------------------------------------------------------------------|----|

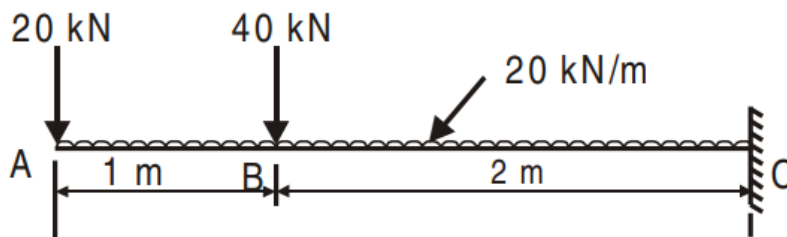


Fig. 2

- |                                                                                                                                                                          |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q.3 (a) A cantilever of length 2 meters fails when a load of 2 kN is applied at the free end. If the section of the beam is 40 mm×60 mm, find the stress at the failure. | 03 |
| (b) Derive using first principle the equation for calculation of maximum shear stress at a section for a beam with rectangular cross section.                            | 04 |

- (c) A wooden beam 100 mm wide and 150 mm deep is simply supported over a span of 4 meters. If the shear force at a section of the beam is 4500 N, find the shear stress at a distance of 25 mm above the neutral axis (N.A.). 07

**OR**

- Q.3** (a) Define: (i) Bending Moment (ii) Point of Contra-flexure 03  
 (b) Draw shear stress distribution diagram for Rectangular, Circular T section and I section. 04  
 (c) A simply supported wooden beam of span 1.3 m having a cross-section 150 mm wide by 250 mm deep carries a point load W at the center. The permissible stresses are  $7 \text{ N/mm}^2$  in bending and  $1 \text{ N/mm}^2$  in shearing. Calculate the safe load W. 07
- Q.4** (a) Define: (1) Centroid, (2) Center of gravity, (3) Center of mass 03  
 (b) State assumptions made in theory of torsion. 04  
 (c) Find the center of gravity of the I-section shown in Fig.3 07

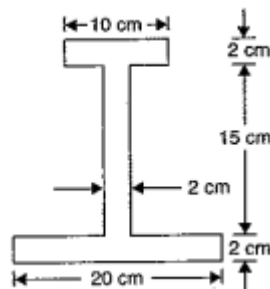


Fig. 3

**OR**

- Q.4** (a) Derive with usual notations the theorem of perpendicular axis. 03  
 (b) Derive torsion equation with usual notations. 04  
 (c) Find the moment of inertia of ISA 100×75×6 about the centroidal XX and YY axes, shown in Fig.4 07

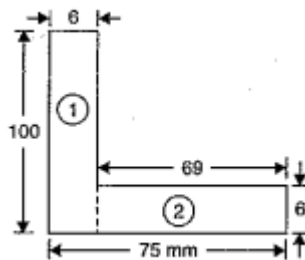


Fig. 4

- Q.5** (a) Explain Stress-Strain diagram for Ductile Material. 03  
 (b) A rod 150 cm long and of diameter 2.0 cm is subjected to an axial pull of 20 kN. If the modulus of elasticity of the material of the rod is  $2 \times 10^5 \text{ N/mm}^2$ , determine: (i) the stress, 04  
 (ii) the strain, and (iii) the elongation of the rod.  
 (c) A metallic bar  $300 \text{ mm} \times 100 \text{ mm} \times 40 \text{ mm}$  is subjected to a force of 5 kN (tensile), 6 kN (tensile), and 4 kN (tensile) along x, y, and z directions, respectively. Determine the change in the volume of the block. Take  $E = 2 \times 10^5 \text{ N/mm}^2$  and Poisson's ratio  $= 0.25$  07

OR

- Q.5** (a) Define stress, strain and poisson ratio. **03**  
(b) A short concrete column  $450 \text{ mm} \times 450 \text{ mm}$  in section is axially loaded to 500 kN. **04**  
The column is strengthened by four, 16 mm diameter steel bars each one at corner. Calculate stresses in concrete and steel. Take  $E_c = 14 \text{ GPa}$  and  $E_s = 210 \text{ GPa}$ .  
(c) A brass bar, having a cross-sectional area of  $1000 \text{ mm}^2$  is subjected to axial forces **07**  
as shown in Fig.5. Find the total elongation of the bar.  
Take  $E = 1.05 \times 10^5 \text{ N/mm}^2$

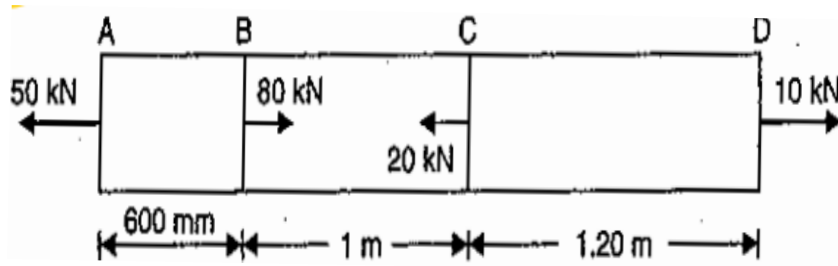


Fig. 5

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Enrollment No./Seat No.:

## GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE01000091/BE01R00091

Date: 29-12-2025

Subject Name: Mechanics of Solids

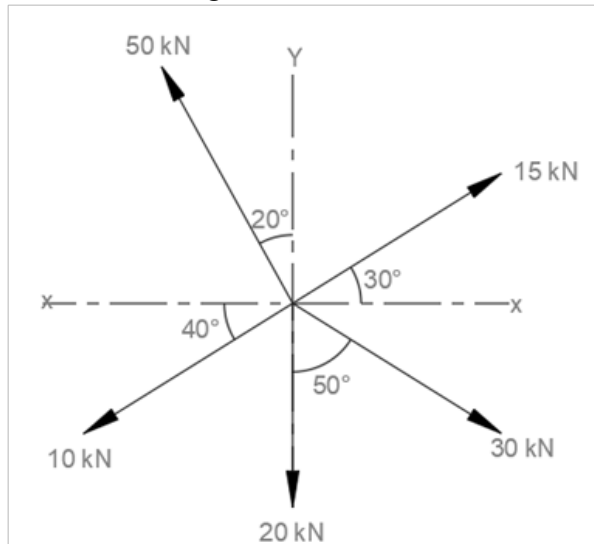
Time: 02:30 PM TO 05:00 PM

Total Marks: 70

### Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

|                                                                                               | Marks     |
|-----------------------------------------------------------------------------------------------|-----------|
| <b>Q.1 (a)</b> Differentiate Statics and Dynamics                                             | <b>03</b> |
| <b>(b)</b> Explain the terms composition of forces and resolution of forces with an example.  | <b>04</b> |
| <b>(c)</b> Find out the magnitude and direction of resultant of force system shown in figure. | <b>07</b> |

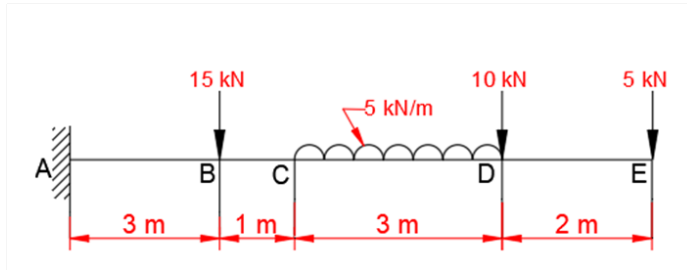


|                                                                                                                                                                                  |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Q.2 (a)</b> State the law of polygon of forces with an example.                                                                                                               | <b>03</b> |
| <b>(b)</b> State and explain Lami's theorem with a diagram.                                                                                                                      | <b>04</b> |
| <b>(c)</b> A simply supported beam 10 m long carries three-point loads at 150 kN, 100 kN and 120 kN at 2m, 4m and 7m from left support. Draw S.F. and B.M. diagram for the beam. | <b>07</b> |

OR



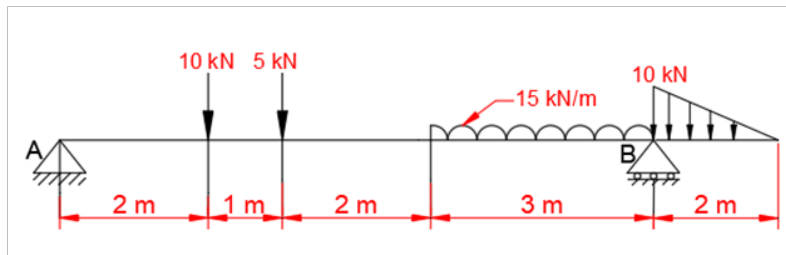
- (c) Determine the support reactions for the beam shown in below Figure. Also Calculate and plot the shear force and bending moment diagrams. 07



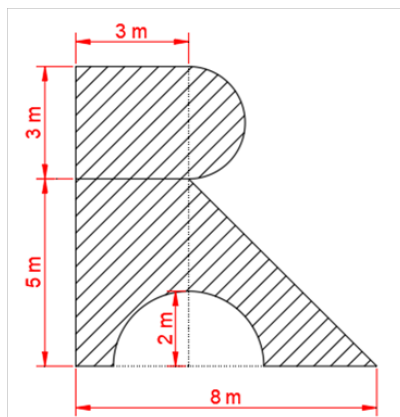
- Q.3 (a) Distinguish between centroid and Center of Gravity. 03  
 (b) Classify different types of beams with neat sketches 04  
 (c) Derive the Equation of moment of Inertia about an axis passing through the Centroid of the rectangle. 07

OR

- (a) Define stress. Also explain types of stresses. 03  
 (b) Determine support reaction for the given beam shown in figure. 04

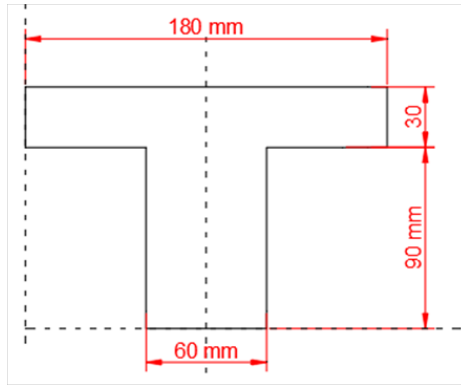


- (c) Determine the centroid of the section shown in Figure below. 07



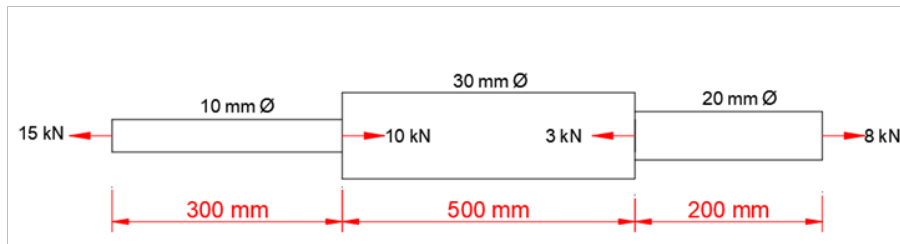
- Q.4 (a) Define Elastic limit and Modulus of Elasticity. 03  
 (b) Derive with usual notations the theorem of perpendicular axis. 04

- (c) Determine moments of inertia of a section shown in figure below about its horizontal centroidal axis. 07



**OR**

- (a) Write the bending equation and explain each term of that equation. 03
- (b) State and prove Pappu's guldinus theorem for surface area of bodies. 04
- (c) A steel bar of 1000 mm long is acted upon by forces as shown in figure . Find the elongation of the bar. Take  $E = 200 \text{ GPa}$ . 07



- Q.5** (a) Draw shear stress distribution diagram for: i) Triangular (ii) Circular and (iii) T - section 03
- (b) Define point of contraflexure. Give an example of where it occurs. 04
- (c) A circular pipe of 100 mm external diameter and 80 mm internal diameter is used as a simply supported beam of 4 m span. Find the safe concentrated load that the beam can carry at the mid-point, if the permissible stress on the beam is  $120 \text{ N/mm}^2$  07

**OR**

- (a) Define Composite bar and Compound bar. 03
- (b) Derive torsion equation with usual notations. 04
- (c) A circular rod of diameter 50 mm and 700 mm long is subjected to a tensile force of 40 kN. The modulus of elasticity for steel is  $200 \text{ kN/mm}^2$ . Find out the stress, strain and elongation of the bar due to applied load. 07

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